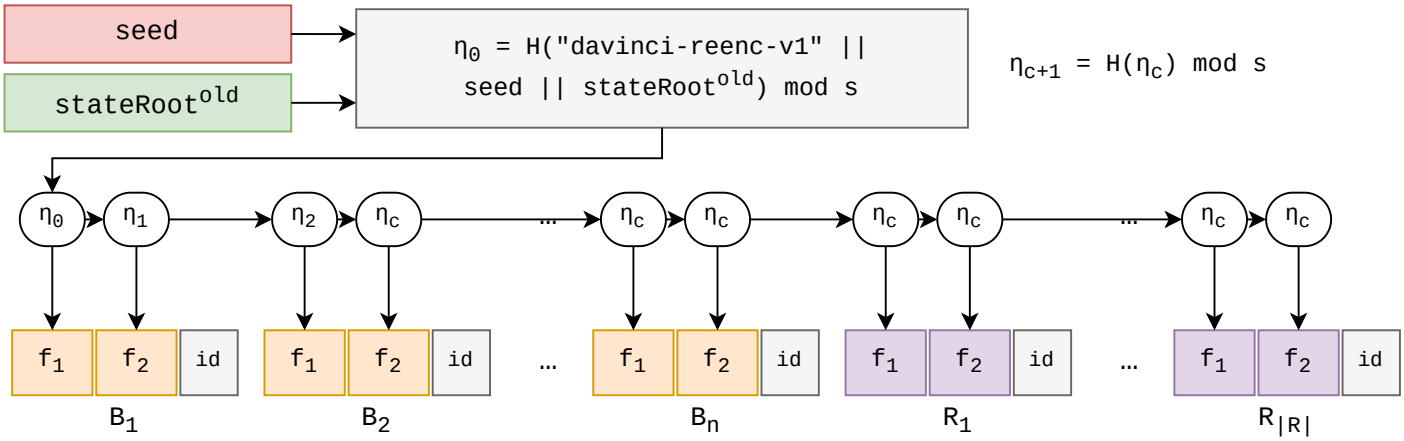


Each active ciphertext consumes the next scalar $\eta_c$; inactive fields (id) must be the identity and consume none (example with $n_f = 2$).

**re-encryption** with $\eta_c$:

$(C_1, C_2) \rightarrow (C_1 + \eta_c \cdot G,$
$C_2 + \eta_c \cdot \text{encryptionKey})$

each chain position is consumed once

**results accumulation**

$A^{new} = A^{old} + \Sigma\ B'_i - \Sigma\ B^{old}_i$
$+ \Sigma\ (R'_j - R_j)$

refreshes add encryptions of zero